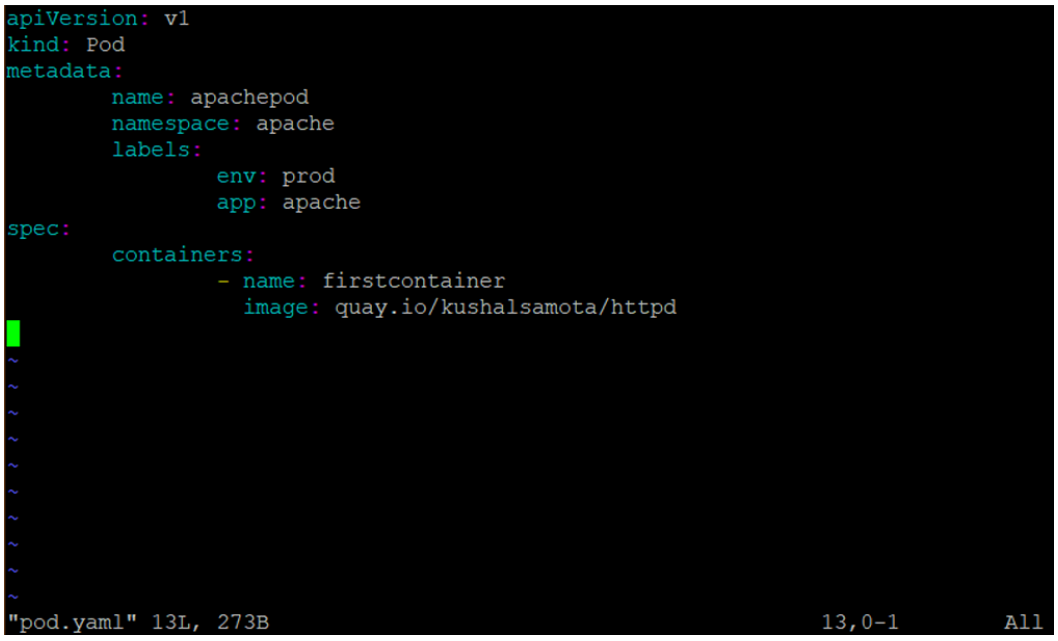
Openshift Exercise
Basic Commands

oc login -u admin -p redhatocp https://api.ocp4.example.com:6443
oc get projects
oc get nodes
oc get nodes -o wide
oc get nodes --show-labels
oc describe node worker01
oc get node worker01 -o yaml > node.yaml
#Open node.yaml and see the content and exit from the file#
oc get pods --all-namespaces
oc get pods -A -o wide
oc get pods -A -o wide --show-labels
oc label node worker01 disk=ssd
oc get nodes --show-labels
oc get nodes -L disk
oc get nodes -l disk=ssd
oc label node worker01 disk-

POD EXERCISE

vim pod.yaml ---- make sure that there are no spaces mistakes.



==========

oc new-project apache
oc create -f pod.yaml
oc get pods -n apache
oc get pods -o wide -n apache
oc get pods --show-labels -n apache
oc describe pod apachepod -n apache
oc logs apachepod
oc get pod apachepod -o yaml -n apache > pod1.yaml
vim pod1.yaml
exit from file
oc delete pod apachepod -n apache
oc get pods -n apache

##

MULTI CONTAINER POD EXERCISE

vim pod.yaml ---- make sure that there are no spaces mistakes.

==========

```
apiVersion: v1
kind: Pod
metadata:
        name: apachepod
        namespace: apache
        labels:
                env: prod
                app: apache
spec:
        containers:
                - name: firstcontainer
                  image: quay.io/kushalsamota/httpd
                - name: secondcontainer
                  image: quay.io/kushalsamota/mysql
                  env:
                          - name: MYSQL_ROOT_PASSWORD
                            value: redhat
~
~
~
~
~
"pod.yaml" 18L, 484B                                          18,0-1        All
```

==========

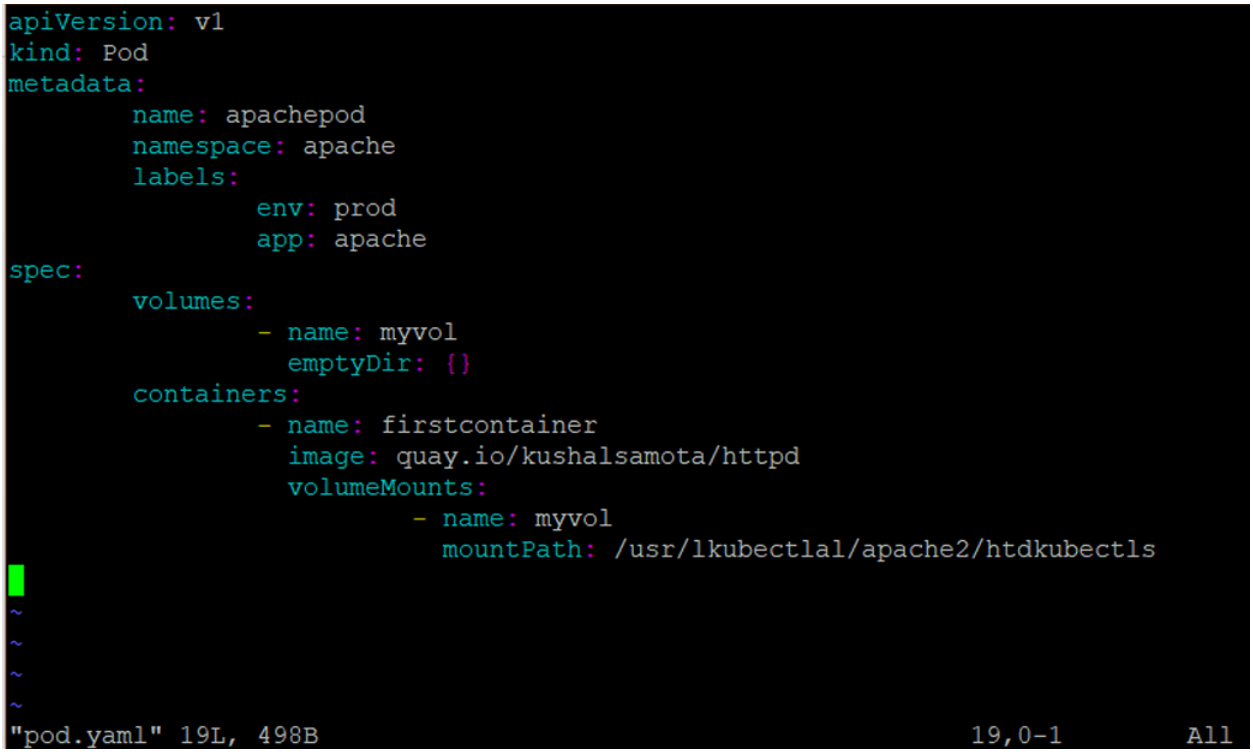
save the file
oc create -f pod.yaml
oc get pods -n apache === two container will be running in a pod
oc get pods -o wide -n apache
oc describe pod apachepod -n apache
oc exec -it apachepod bash -n apache
exit
oc exec -it apachepod -c secondcontainer bash -n apache
exit
oc logs apachepod -c firstcontainer -n apache
oc logs apachepod -c secondcontainer -n apache
oc delete pod apachepod -n apache
oc get pods -n apache

##########

POD WITH VOLUME(EMPTY DIR) EXERCISE

vim pod.yaml ---- make sure that there are no spaces mistakes.

==========



==========

oc create -f pod.yaml
It will give you the error as the mount path is wrong.
Edit the pod.yaml again and change the mount path to /usr/local/apache2/htdocs
oc apply -f pod.yaml
oc get pods -n apache
oc describe pod apachepod -n apache
oc delete pod apachepod -n apache

##########

REPLICA SET EXERCISE

==========

vim rs.yaml ---- make sure that there are no spaces mistakes.

==========

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: mysql
  namespace: apache
spec:
  replicas: 5
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      name: mysqlpod
      labels:
        app: mysql
    spec:
      containers:
      - name: php-redis
        image: quay.io/kushalsamota/mysql
        env:
          - name: MYSQL_ROOT_PASSWORD
            value: redhat
~
~
~
~
~
~
                                                                23,0-1        All
```

==========

save the file

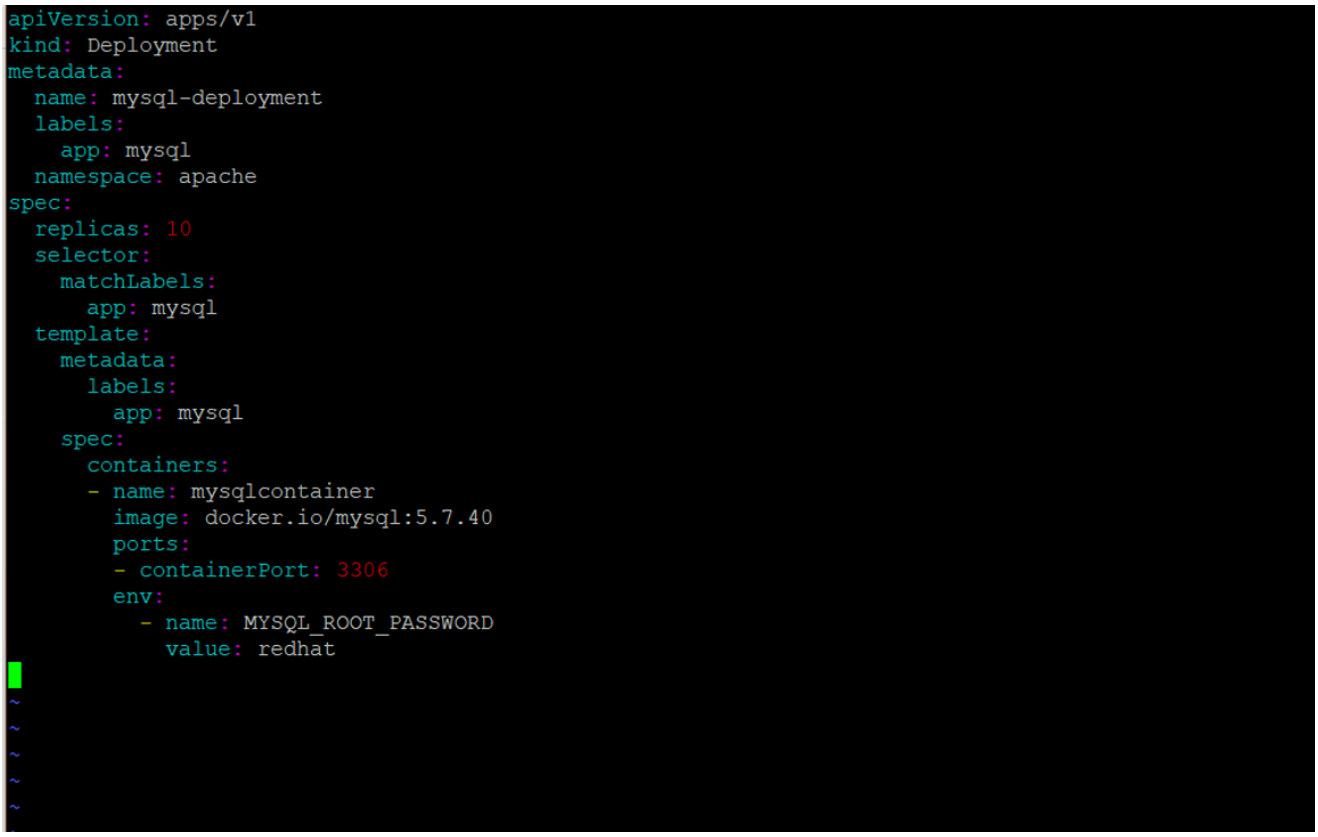
==========

oc create -f rs.yaml
oc get all -n apache
oc scale rs/mysql --replicas=5 -n apache
oc get pods -n apache
oc scale rs/mysql --replicas=1 -n apache
oc get pods -n apache
oc delete rs mysql -n apache
oc get all -n apache

##

DEPLOYMENT EXERCISE

vim dep.yaml ---- make sure that there are no spaces mistakes.



```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mysql-deployment
  labels:
    app: mysql
  namespace: apache
spec:
  replicas: 10
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
      - name: mysqlcontainer
        image: docker.io/mysql:5.7.40
        ports:
        - containerPort: 3306
        env:
          - name: MYSQL_ROOT_PASSWORD
            value: redhat
~
~
~
~
~
```

oc create -f dep.yaml
oc scale deployment/mysql-deployment - -replicas=3 -n apache
oc get all
#It might give you an error as the images from docker.io gives pull rate limit error.
oc delete deployment mysql-deployment
#Open dep.yaml file and change the image path to quay.io/kushalsamota/mysql

oc apply -f dep.yaml
oc get all -n apache
oc set image deployment mysql-deployment mysqlcontainer=quay.io/kushalsamota/mysql -record -n apache
oc rollout history deployment mysql-deployment -n apache
oc get all -n apache
oc rollout history deployment mysql-deployment -n apache
oc get all -n apache
oc rollout undo deployment mysql-deployment --to-revision=1 -n apache
oc rollout history deployment mysql-deployment -n apache
oc delete deployment mysql-deployment -n apache

##

SECRET EXERCISE

vim dep.yaml ---- make sure that there are no spaces mistakes.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mysql-deployment
  namespace: apache
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
      - name: mysql
        image: quay.io/kushalsamota/mysql
        env:
                - name: MYSQL_ROOT_PASSWORD
                  valueFrom:
                          secretKeyRef:
                                  name: mysqlpassword
                                  key: passwordofmysql
~
```

oc create secret generic mysqlpassword --from-literal=passwordofmysql=redhat –n apache
oc create -f dep.yaml
oc get pods –n apache
oc describe deployment mysql-deployment -n apache
oc describe secret mysqlpassword -n apache
oc delete deployment mysql-deployment -n apache

oc delete secret mysqlpassword -n apache

##

CONFIGMAP EXERCISE

vim dep.yaml ---- make sure that there are no spaces mistakes.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mysql-deployment
  namespace: apache
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
      - name: mysql
        image: quay.io/kushalsamota/mysql
        env:
                - name: MYSQL_ROOT_PASSWORD
                  valueFrom:
                          configMapKeyRef:
                                  name: mysqlpassword
                                  key: passwordofmysql
~
~
```

oc create configmap mysqlpassword --from-literal=passwordofmysql=redhat –n apache
oc get configmap -n apache
oc get pods –n apache
oc describe configmap mysqlpassword -n apache
oc create -f dep.yaml
oc describe deployment mysql-deployment -n apache
oc delete deployment mysql-deployment -n apache
oc delete configmap mysqlpassword -n apache
oc delete project apache